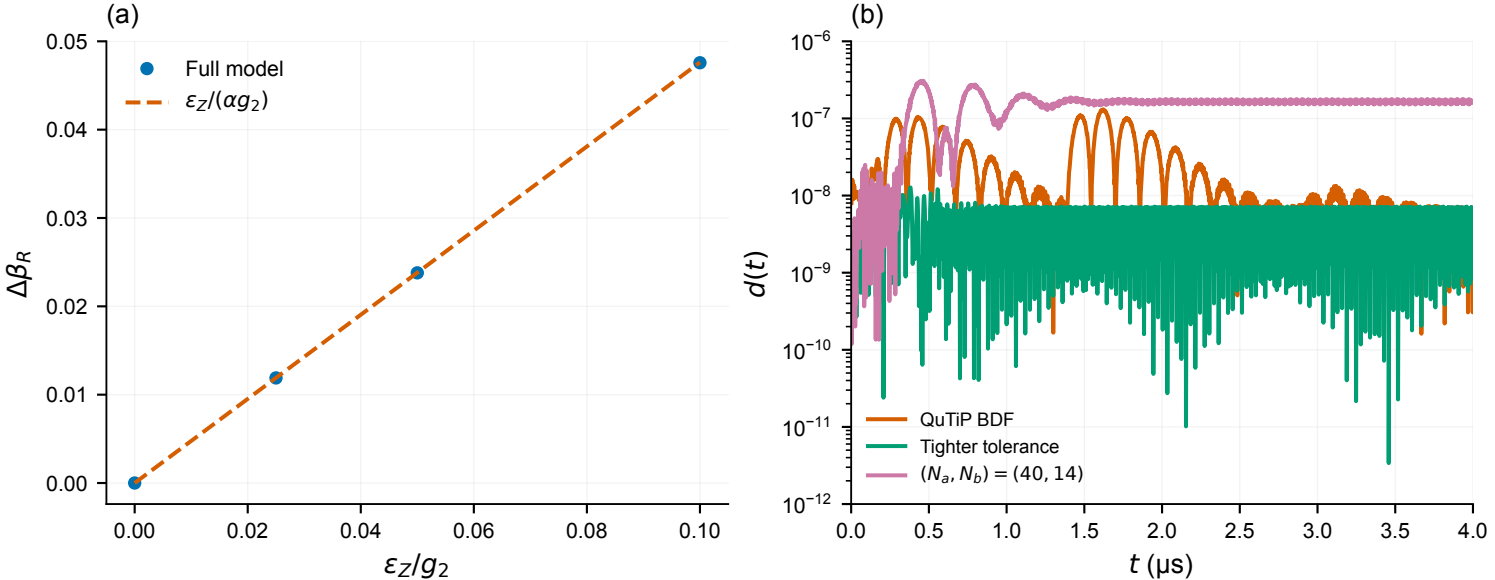

(a)
Full model
$\varepsilon_Z/(\alpha g_2)$
$\Delta\beta_R$
$\varepsilon_Z/g_2$
0.05
0.04
0.03
0.02
0.01
0.00
0.00
0.02
0.04
0.06
0.08
0.10
(b)
$d(t)$
$t$ (μs)
QuTiP BDF
Tighter tolerance
$(N_a, N_b) = (40, 14)$
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
$10^{-10}$
$10^{-11}$
$10^{-12}$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0